

[This question paper contains 8 printed pages.]

Sr. No. of Question Paper : 6118

Your Roll No.

L

Unique Paper Code : 2344002004

Name of the Paper : Introduction to Web Programming

Name of the Course : Computer Science Generic Elective

Semester : IV

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Section A is compulsory.
3. Attempt any four questions from Section B.
4. Parts of a question must be answered together.

Section- A

1. (a) Write an HTML program to display the message "Welcome to my Webpage" using the following formatting specifications:

Font size: 30px

Font family: Arial

Font color: yellow

(3)

- b) Write HTML-CSS code to design a webpage with the following list. (3)

I Web Development

- HTML
- CSS

II. Programming

- C++

(3)

- Python

- (c) Write a JSON object to store details of a student including name, age, and marks in three subjects. (3)

(3)

- (d) Consider the following HTML and CSS code:

```
<html>
```

```
<head>
```

```
<style>
```

```
p { color: blue; }
```

```
.text { color: green; }
```

```
#id1 { color: red; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<p id="id1" class="text">Sample Text</p>
```

```
</body>
```

```
</html>
```

- (i) What will be the final color of the paragraph?

- (ii) Name the selector responsible for this color.

- (iii) Why is this selector applied instead of the others?

- (e) Differentiate between HTML tags and HTML attributes. Give two examples of each. (3)

- (f) What is an array in JavaScript? Declare an array of five fruits and print the third fruit using Javascript. (3)

- (g) What is the difference between id and class selectors in jQuery? (3)
- (h) Create a webpage using HTML that performs the following: (3)
- (i) Display a heading and a paragraph
 - (ii) Creates a hyperlink that opens in a new tab
 - (iii) Insert a horizontal rule

(i) Write a JavaScript program to change the background color of a webpage when a button is clicked. (3)

(i) Explain the purpose of the following tags in HTML: (3)

(i) `<a>`

(ii) `<img>`

(iii) `<br>`

Section-B

2/ (a) Write a JavaScript program that takes a number as input from the user and determines whether it is positive, negative, or zero. If the number is positive and odd, the program should display its multiplication table up to 10. If the number is positive and even, it should calculate and display the sum of the first N natural numbers, where N is the number entered by the user. If the number is negative, it should display an alert message stating "Negative number entered". (8)

(b) Explain the CSS Box Model in detail. Describe each of its components and discuss how padding, border, and margin affect the layout of an element. Support your answer with a neat diagram and a suitable CSS example. (7)

3. Compare and explain the following javascript event handlers based on their functionality: (9)

(i) onclick vs ondblclick

(ii) onmouseover vs onmouseout

(iii) onkeydown vs onkeyup

- (b) Explain rowspan and colspan attributes in HTML table. Design a webpage using HTML to display the following table. Also change the background color and text color of table using CSS. (6)

Day	Period 1	Period 2	Period 3
Monday	Math	English	Science
	History and Art		Music

4. Differentiate between the following: (8)

(a) block and inline elements

(b) and

(c) Web browser and web server

(d) margin and padding

- (b) Create a webpage that accepts employee details such as ID, Name, and Department through input fields. Using JavaScript, display the entered details dynamically in an HTML table when the user clicks a button. (7)

2:30
4:30

and

5. (a) Create an HTML page using `<div>` element that divides the browser window into two horizontal sections: the top section should display "My Webpage" and the bottom section should be further divided vertically into a left section showing "Table of contents" and a right section showing "Main Content". (8)
- (b) What is jQuery? Find and correct the errors in the following jQuery code: (7)

```
<script>
$(document).ready(function(){
    $("button").onlick(function {
        $("#msg").html = "Button Clicked"
        $(".box").css("color" "blue");
        if($("#input").val == ""){
            alert("Empty");
        }
    });
});
</script>
```

6. (a) Write a complete HTML and JavaScript program to create a simple calculator that accepts two numbers from the user through input fields and performs addition, subtraction, multiplication, or division based on the operation selected from a dropdown menu. Display the result in a text box. Also include a clear button to reset all input and output fields. (9)

Number 1:

Number 2:

Operation:

Result:

(b) Write the output of the following code and justify your answer. (3+2+2=7)

(i) `<!DOCTYPE html>`

`<html>`

`<body>`

`<script>`

`var a = 5;`

`var b = 2;`

`document.write(a + b);`

`document.write("<br>");`

`document.write(a + "2");`

`document.write("<br>");`

`document.write(a - "2");`

`</script>`

`</body>`

`</html>`

(ii) `<!DOCTYPE html>`

`<html>`

`<body>`

`<p id="demo"></p>`

`<script>`

`var x = 10;`

`var y = "10";`

`if (x == y) {`

`document.getElementById("demo").innerHTML = "Equal";`

`} else {`

`document.getElementById("demo").innerHTML = "Not Equal";`

```
}  
</script>
```

```
</body>
```

```
</html>
```

(iii) `<!DOCTYPE html>`

```
<html>
```

```
<body>
```

```
<p id="demo"></p>
```

```
<script>
```

```
var x = "";
```

```
if (x) {
```

```
    document.getElementById("demo").innerHTML = "True";
```

```
} else {
```

```
    document.getElementById("demo").innerHTML = "False";
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

7-1-26

7. (a) Describe the three types of Cascading Style Sheets (CSS), namely Inline CSS, Internal CSS, and External CSS. Illustrate their usage by creating a suitable webpage that applies each type of CSS to different elements. Also, compare these three types of CSS with respect to their scope, reusability, and suitability in web development. (8)

- b) Design an HTML webpage that includes a user registration form with fields for Name, Email, Password, and Mobile Number. Implement JavaScript

form validation to ensure that all required fields are properly filled before submission. The validation should check that the Name field is not empty, the Email follows a valid format (such as example@gmail.com), the Password contains at least 6 characters, and the Mobile Number consists of exactly 10 digits. Appropriate error messages should be displayed for invalid inputs, and the form should not be submitted until all the validation conditions are satisfied. (7)